

120 V, 155 A Traction Inverter Using Eight Parallel GaN HEMTs

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Abstract—A six-phase, GaN-based inverter using 8 parallel connected EPC2304 200 V 5 mΩ GaN HEMTs per switch is presented. The design is configured as two identical three-phase inverters with the option of a shared DC bus to enable recirculation of power. The design is used to power two mechanically coupled PMSMs in a dynamometer test rig. The use of GaN transistors enables the design to operate up to frequencies of 100 kHz. Static and dynamic current sharing of the GaN HEMTs is ensured by careful consideration of symmetry within the gate loop layout, and is verified using thermal imaging of the GaN HEMTs. Each three-phase inverter can reach an output power of 22.8 kW, with estimated HEMT temperatures not exceeding 70°C at 50 kHz switching frequency.

Index Terms—Gallium Nitride, HEMTs, Paralleling, Traction Inverter, dv/dt, FEA, Wide Bandgap Devices, Thermal Imaging

I. INTRODUCTION

Wide bandgap devices offer favourable electrical and thermal performance compared to Si devices making them an attractive replacement in EVs [1]. The introduction of SiC-based inverters in commercial EVs has already led to increased switching frequencies compared to older Si-based designs, such as the Tesla Model 3 at 10 kHz in 2017 [2] versus the Nissan Leaf at 5 kHz in 2011 [3]. GaN offers a further improvement over SiC due to superior characteristics such as lower specific on-state resistance, and lower gate and output capacitance for comparable device ratings [4]. Under similar conditions, 650 V GaN HEMTs have been shown to exhibit x3.2 lower switching losses than SiC MOSFETs [5], enabling higher switching frequencies [6] and reduced passive component size that are highly desirable in EV applications. Although GaN HEMTs can offer improvements in efficiency and power density in EV inverters, there are still implementation challenges related to thermal and parasitic problems, such as high dv/dt reducing motor lifetimes [7], that limit commercial realisation of GaN-based inverters in EVs [8].

The authors could not find any commercially available GaN-based traction inverters with the necessary flexibility to enable a full range of testing in a motor drive context. Therefore, the

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custom design of a dual three-phase inverter was undertaken. This necessitated the design of a high current half bridge module as described in this paper.

By having the test motor mechanically coupled to a second motor and inverter in a dynamometer (dyno) rig, power can be recirculated back into the DC link of the inverter. This means that the power supply need only provide power for losses in the system, greatly increasing the possible power rating of inverter and motor that can be tested in a laboratory setting. This is in contrast to using a passive dyno, where the power provided by the motor is dissipated into the environment. Fig. 1 shows an electrical schematic of the motor dyno test rig.

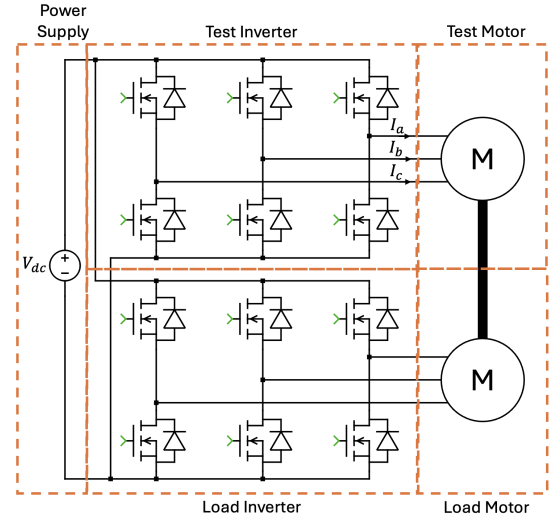


Fig. 1: Electrical schematic for the testing of the dual, three-phase inverters and motors.

The rig is composed of two Plettenburg Nova 15 water cooled motors with their shafts coupled, as well as a 3D printed housing and coolant temperature sensors for calorimetric measurements of motor losses.

The current handling capability of individual GaN HEMTs is constrained by their device structure, limiting achievable die size and maximum drain current [9]. In high current applications such as EV traction inverters, paralleling of multiple GaN HEMTs per switch is necessary.

Parallel connected GaN HEMTs should share current equally to prevent the unequal heating of one particular HEMT, which increases probability of failure of said HEMT [10]. By matching the gate loop impedances, the turn-on and turn-off of the HEMTs should occur simultaneously, ensuring equal switching losses [11]. It is also important that the layout of the half bridge promotes good steady state current sharing by balancing the current pathways of each HEMT such that the total resistance of each pathway is equal.

Table I summarises previous publications [12]–[28] investigating the paralleling of GaN HEMTs. The table is ordered by the most complex configuration tested. Eq. 1 details how the Device Power Utilisation (DPU) is computed.

$$DPU = \frac{V_{dc} I_{ph}}{n V_{ds}^2 / R_{ds(on)}} 10^4 \quad (1)$$

V_{ds} is the rated drain-source voltage, and $R_{ds(on)}$ is the on-state resistance of the HEMT. n is the number of HEMTs in parallel. V_{dc} is the DC link voltage and I_{ph} is the highest tested phase current of the test configuration.

It is evident that the majority of existing work has been limited to double pulse testing or simple converter topologies, with only a limited number of studies progressing to the full testing of GaN power modules in continuous operation in a three-phase system with a motor acting as a load. No GaN paralleling studies were found which had an active dyno test rig recirculating power through a second three-phase inverter.

II. DESIGN

A motherboard PCB mounts six half bridge power modules, organised into two three-phase inverters. The motherboard also houses the bulk DC link capacitors and current sensors for each phase and the DC links. The system is controlled by an STM32F410RB microcontroller. The modules attach to the motherboard using an array of pin header connectors. Fig. 2 shows the complete GaN-based inverter, including all six modules, and the populated motherboard PCB.

Fig. 3 is a electrical schematic detailing one of the modules in the three-phase inverter. Each half bridge power module is composed of a separate half bridge and gate driver PCB. This is to help facilitate fault isolation, debugging, and ease modification of the PCBs.

A. Half Bridge PCB Design

The half bridge PCB is composed of four layers, with the outside layers being 4 oz copper, and the inside layers being 2 oz copper. The top side of the half bridge PCB is used to house the main bulk of the components including GaN HEMTs, DC link capacitors, NTC thermistor, and copper pours for the V_{dc} , PGND, and switch node. The bottom side has all the pin headers for power and signals from the motherboards and the HEMT gate and source connections for the gate driver PCB, and a switch node copper pour. The middle layers of the PCB contain copper pours which complete the switched current commutation loop (SCCL).

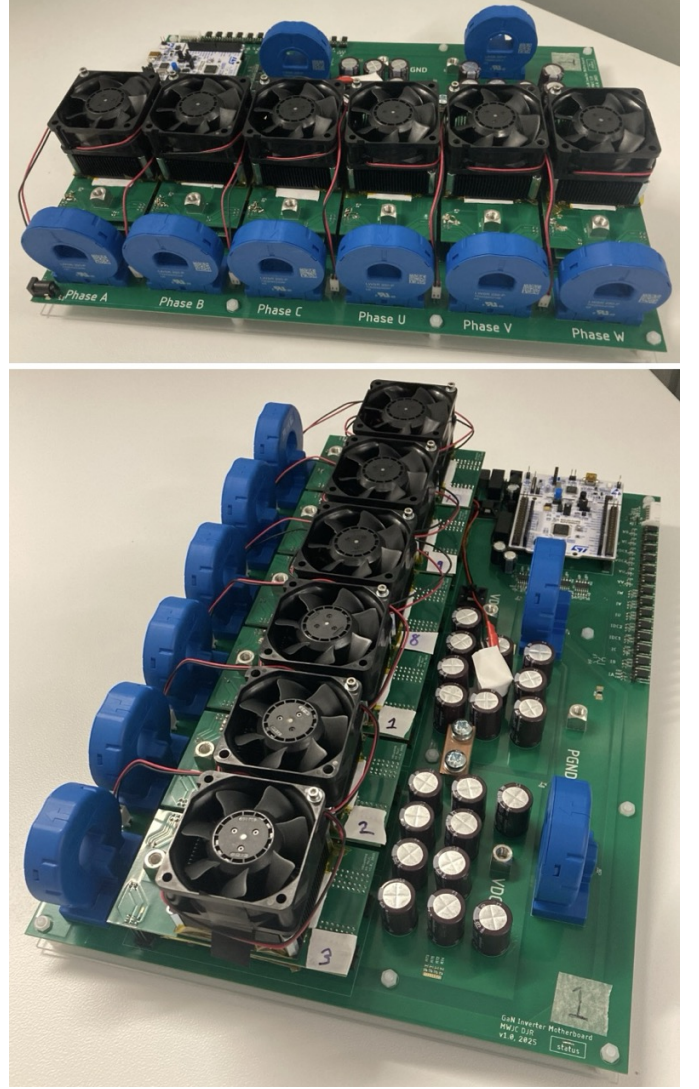


Fig. 2: Two photos of the complete GaN-based inverter.

As shown in fig. 4, the HEMTs are organised into two straight lines with pin headers connecting to the gate driver PCB placed right at the gate and source connections of the HEMTs to minimise gate loop length. Two rows of 18 MLCCs are used to create a local DC link capacitance of $8.36 \mu\text{F}$ on the half bridge PCB, close to the HEMTs. An array of 32 100 mil pin headers are used to connect the V_{DC} and PGND from the motherboard to the half bridge PCB. This method of connection between the half bridge PCB and motherboard PCB is chosen to allow easy connection and disconnection of the power modules, as well as allowing sufficient spacing between the PCBs to make space for all the parts of the power module which hang below the half bridge PCB itself. Terminal blocks are press-fitted into the PCB to act as cable connection terminals for the switch node of each half bridge.

Fig. 4 shows the top side of the half bridge PCB, including key sizes and components in the design.

Fig. 5 is a cross-section of the half bridge PCB, including the two switched current commutation loop pathways.

TABLE I: Summary of paralleling GaN HEMT literature.

Reference	Parallel Device Used	Device Rating $V_{ds}/I_d/R_{ds(on)}$	Test Configuration	Test Rating $V_{dc}/I_{ph}/f_{sw}$	DPU
[12]	x2 GS-065-060-2-D2	650 V / 60 A / 25 mΩ	Simulation	400 V / 100 A	11.8
[13]	x4 GS61008	100 V / 90 A / 7 mΩ	Double pulse test	48 V / 367 A	30.8
[14]	x4 GS66516T	650 V / 60 A / 25 mΩ	Double pulse test	400 V / 280 A	16.6
[15]	x2 GS-065-15V-7-D6	650 V / 150 A / 10 mΩ	Double pulse test	350 V / 25 A	1.0
[16]	x4 GS66508T	650 V / 30 A / 50 mΩ	Double pulse test	400 V / 20 A	2.4
[17]	x2 GS-065-150-1-D	650 V / 150 A / 10 mΩ	Double pulse test	450 V / 150 A	8.0
[18]	x5 GS66508P	650 V / 30 A / 50 mΩ	Double pulse test	350 V / 100 A	8.3
[19]	x2 GS66516B	650 V / 60 A / 25 mΩ	Double pulse test	400 V / 80 A	9.5
[19]	x4 GS66516T	650 V / 60 A / 25 mΩ	Double pulse test	400 V / 160 A	9.5
[20]	x12 GS65516T	650 V / 60 A / 25 mΩ	DC current applied externally	290 A injected	8.1
[21]	x4 GS66508B	650 V / 30 A / 50 mΩ	Boost converter operation	100 V to 400 V / 1900 W	0.6
[22]	x3 V08TC6551A2	650 V / 200 A / 7.8 mΩ	Three-phase H-bridge	400 V / 260 A / 16 kHz	6.4
[23]	x2 GS66518B	650 V / 60 A / 50 mΩ	Passive three-phase RL load	550 V / 10.2 A / 70 kHz	3.3
[24]	x8 CoolGaN™ HEMT	100 V / 75 A / 3 mΩ	Passive three-phase RL load	48 V / 125 A / 50 kHz	2.3
[25]	x4 GS61008P	100 V / 90 A / 7 mΩ	Passive three-phase RL load	48 V / 113 A / 20 kHz	9.5
[26]	x4 EPC2032	100 V / 101 A / 3 mΩ	Single motor operation	60 V / 150 A / 100 kHz	6.8
[27]	x2 EPC2065	80 V / 60 A / 2.7 mΩ	Single motor operation	48 V / 35.4 A / 100 kHz	3.6
[28]	x4 V08TC065S1X11	650 V / 200 A / 8 mΩ	Single motor operation	400 V / 350 A	6.6
This work	x8 EPC2304	200 V / 133 A / 5 mΩ	Dual motor operation	120 V / 155 A / 100 kHz	2.9

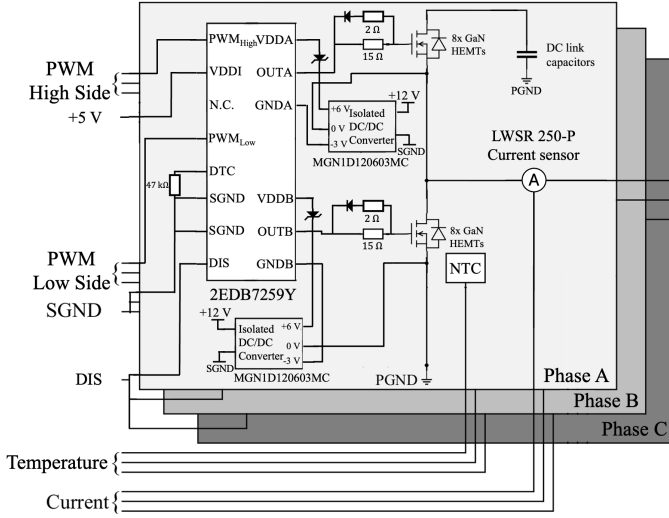


Fig. 3: Electrical schematic detailing one phase of the inverter.

The inductance of a PCB trace can be estimated using eq. 2 [25] and can be used to calculate an approximation of the SCCL inductance, assuming that skin, proximity, and fringing effects are negligible:

$$L_{SCCL} = \frac{\mu_r \mu_0 h l}{w} \quad (2)$$

μ_r is the relative permeability of the PCB substrate, μ_0 is the permeability of free space, h is the height of the trace, l is the length of the trace, and w is the width of the trace. The estimated inductance of the SCCL illustrated by the blue path in fig. 5 is 0.74 nH, and the inductance of the red path is 0.34 nH.

Once the half bridge and gate driver PCBs are fully populated they can be assembled into a full half bridge module, which includes the heat sink, fan, TIM, and backing plate with springs. Fig. 6 is a picture of one of the complete half bridge modules.

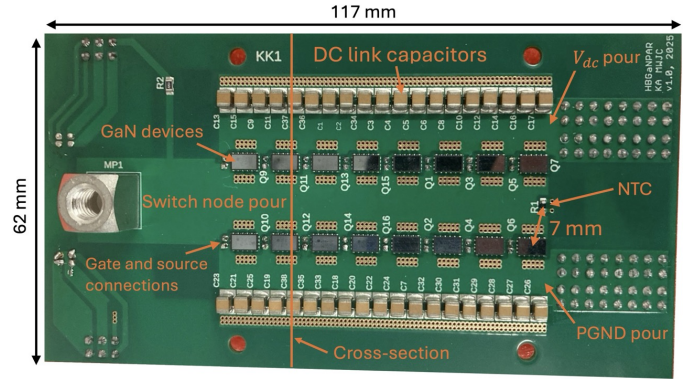


Fig. 4: The top side of the half bridge PCB, including GaN HEMTs, DC link capacitors, and thermistor. The orange vertical line indicates the cross-section shown in fig. 5.

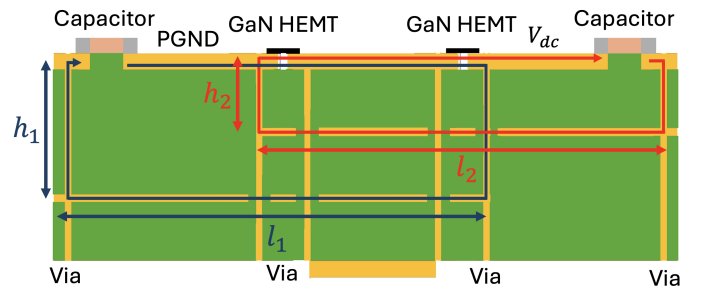


Fig. 5: Cross-section of the half bridge PCB, with the two SCCLs paths shown in red and blue. Note that sizing of the cross-section is not to scale.

The TIM used is a 1.5 mm thick A1780 TIM pad, which has a thermal conductivity of 17.8 W/mK. Mounted onto that is a 60 x 60 mm aluminium heat sink with a thermal resistance of 0.9 K/W when forced convection is used. Springs are used to apply a compressive force of 9 N per spring to the heat sink to ensure good thermal contact between the HEMT and TIM, and the TIM and heat sink. The thickness of the TIM

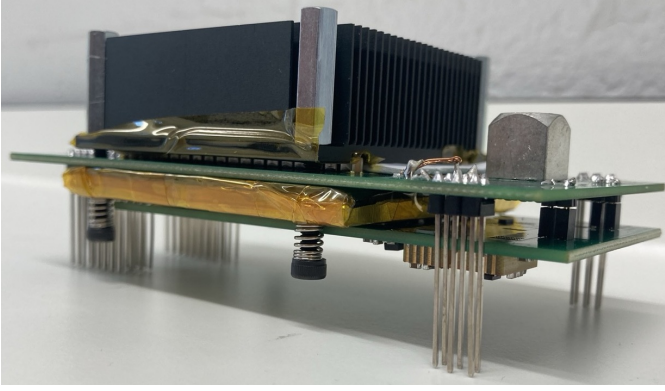


Fig. 6: A complete half bridge module including the half bridge and gate driver PCBs, Heat sink, baseplate, and TIM. The cooling fan is not shown.

directly above the HEMTs is 0.8 mm thick. A backing plate on the bottom side of the half bridge PCB is used to prevent warping of the PCB that could occur due to the localised force from the springs. Fig. 7 is a cross-sectional diagram of the half bridge module, visualising the key layers in the stack up.

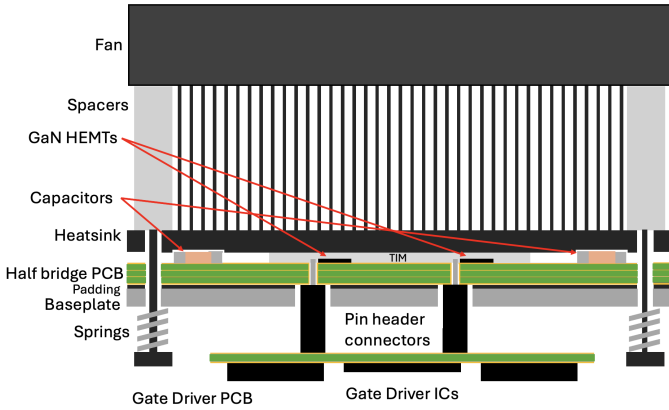


Fig. 7: A cross-sectional diagram of the complete half bridge power module, containing both the half bridge and gate driver PCBs.

A 10 k Ω NTC thermistor is used to monitor the temperature of the GaN HEMTs on the half bridge PCB. The thermistor is placed 7 mm from the closest HEMTs, and both are embedded in TIM, creating good thermal coupling between the thermistor temperature and the HEMT junction temperature.

B. Gate Driver PCB Design

The gate driver PCB is composed of two layers of 1 oz copper. The top side of the gate driver PCB is used to house the main bulk of the gate driver components, and the bottom side is used for the sink and source gate traces.

To encourage symmetric switching of the parallel GaN HEMTs, the impedances of their gate loops should be matched as close as possible. To match the gate loop impedances, the physical layout of the traces is designed such that there is symmetry in each pathway, similar to designs in previous publications [24], [25]. Fig. 8 shows the PCB schematic of

the bottom layer of the gate driver PCB. Included are the sink and source traces and gate resistors, and the gate and source pin connections, for the high and low side HEMTs. The source return path is a copper pour on the top layer of the gate driver PCB, not shown in this figure.

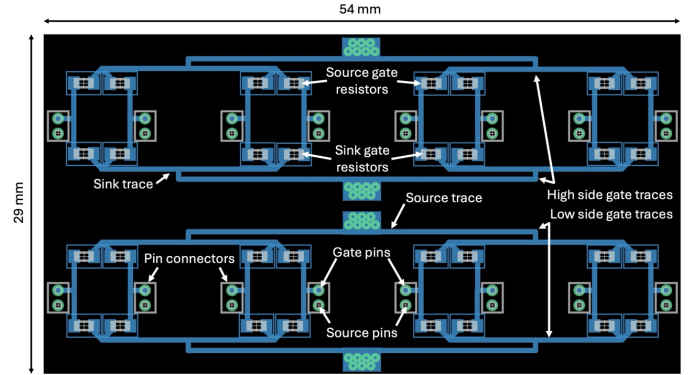


Fig. 8: Bottom layer of the gate driver PCB schematic, showing the sink and source gate loop traces for the top and bottom side HEMTs, as well as other key details.

An Infineon 2EDB7259Y is selected to be the gate driver due to its large current capability of 5 A source and 9 A sink. The sink and source resistors are selected to keep the peak sink and source current below the rating of the gate driver. Because the gate driver has a single output pin for both sinking and source current, a schottky diode is used to reduce the impedance of the sink pathway. The selected gate resistances are 15 Ω for the source gate resistor and 2 Ω for the sink gate resistor. Pin header connectors are used to individually connect the gate driver to the gate and source pins of each HEMT on the half bridge PCB.

Isolated DC/DC Converters are used to provide a bipolar voltage of +6V/-3V to the top and bottom gate driver channels. A zener diode is used to reduce the positive voltage from the power supplies of 6 V down to 5.3 V to increase the safety margin between the positive V_{gs} applied to the HEMT and their data sheet rating. The selection of a bipolar power supply is to guard against the false turn-on of any of the HEMTs [29], which could be caused by high dv/dt and parasitics [30] in the design coupled with the fact that GaN HEMTs have a low threshold voltage.

Fig. 9 shows the top side of the gate driver PCB.

C. Estimating HEMT Temperature

Figure 4 shows the distance between the thermistor and the HEMTs on the top side of the half bridge PCB. The thermistor will be colder than the HEMTs and so using the MATLAB Partial Differential Equation Toolbox, a steady state thermal simulation of the system is conducted to model the difference between the HEMTs junction temperature and the temperature measured by the thermistor. The HEMTs are modelled as each generating equal thermal power loss, and it is assumed that all heat in the system dissipates via the heat sink on top, as this is the dominant thermal pathway to ambient.

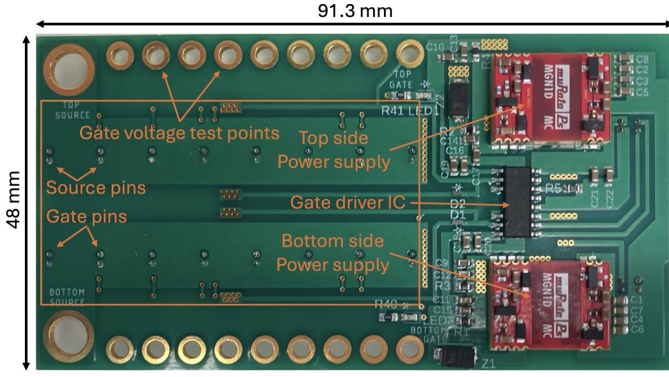


Fig. 9: The top side of the gate driver PCB. The orange box is the area occupied on the bottom side by the traces and components shown previously in fig. 8.

To simplify the geometry of the heat sink, a simplified heat sink model is used, which is composed of an aluminium plate of the same thickness as the base of the heat sink, and then a convective boundary condition with a heat transfer coefficient chosen to give a resulting thermal resistance equal to the value given in the heat sinks' data sheet of 0.9 K/W. This simplification retains the key thermal characteristics of the heat sink: The magnitude of the thermal resistance to ambient is maintained by enforcing a specific heat transfer coefficient at the boundary, and the horizontal thermal resistance across the heat sink base.

Fig. 10 shows the geometry of the system modelled, including the HEMTs, thermistor, TIM, and simplified heat sink. Ambient temperature is set to 25 °C.

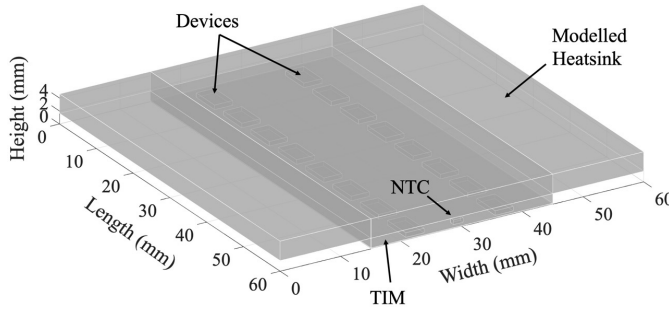


Fig. 10: Geometry of the modelled thermal system.

Fig. 11 shows the temperature distribution of a horizontal plane halfway through the TIM, HEMTs, and thermistor, for a FEA thermal simulation where the HEMTs are dissipating 3 W each. Note there is a 1.2 °C temperature difference between the hottest and coldest HEMTs, which is due to the asymmetric placement of the heat sink over them.

As the thermal system modelled is linear, there is a fixed error of 10.1% between the increase in temperature of the thermistor and the temperature rise of the HEMT, which can be expressed as,

$$T_J - T_{amb} = \alpha(T_{thermistor} - T_{amb}) ; \text{ where } \alpha = 1.101 \quad (3)$$

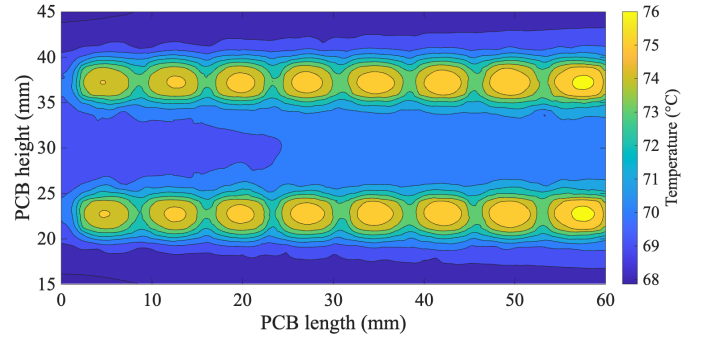


Fig. 11: Temperature distribution of a cross section halfway through the TIM, HEMTs, and thermistor.

Measurement of the thermistor temperature is used in section III to estimate the rating of the half bridge power modules. The mean thermal resistance of the HEMTs is 16.9 K/W, meaning that for a average HEMT junction temperature of 100 °C, each HEMT can dissipate 4.4 W of heat.

III. EXPERIMENTAL VERIFICATION AND RESULTS

A. Switching Behaviour

Fig. 12 shows the V_{gs} waveforms during turn-on and turn-off of a top and bottom HEMT when not conducting.

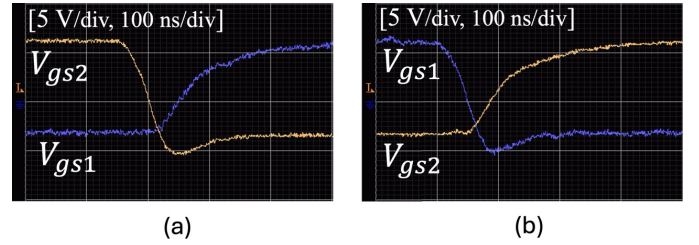


Fig. 12: V_{gs} of a top and bottom side HEMT during (a) turn-on, and (b) turn-off.

Fig. 13 shows the rising and falling switch node voltage transition waveforms when not conducting. The scale of the waveform image is 5 ns/div and 20 V/div.

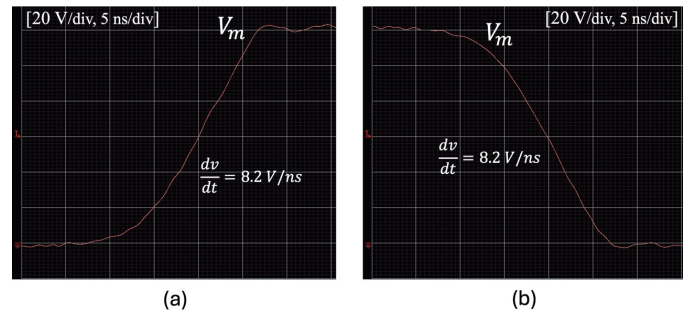


Fig. 13: Switch node voltage waveforms during (a) turn-on, and (b) turn-off.

The switch node voltage turn-on and turn-off have a dv/dt of 8.2 V/ns. This is higher than what is usually achieved by Si (3 V/ns [31]), and is above limits set by standards for

industrial inverter systems (3-6 V/ns [32]), but is less than what is projected for next-generation electric motors dv/dt ratings (10-15 V/ns [33])

B. Full Bridge Converter Test

To verify operation at the rated output current, voltage, and switching frequency of the inverter, two half bridge power modules are operated in a full bridge setup with an EA-EL 9080-200 acting as an electrical load as shown in fig. 14. The duty cycle of the half bridges is set to $50\% \pm 5\%$ such that a current flows from one half bridge to the other. An output filter is used to ensure that the high dv/dt doesn't interfere with the electronic load's current control. Table II contains the results from two experiments conducted on the power modules

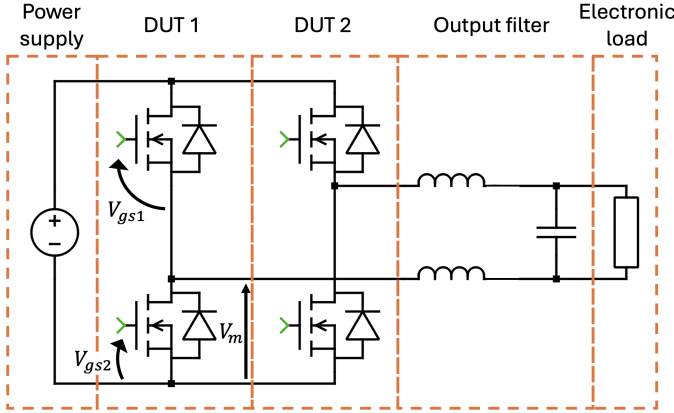


Fig. 14: Electrical schematic for the testing of two half bridge power modules in a full bridge arrangement.

TABLE II: Experimental Results of the Full Bridge Power Test

Parameter	50 kHz Test	100 kHz Test
DC Link Voltage		120 V
Phase Current		155 A
Switching Frequency	50 kHz	100 kHz
Maximum thermistor Temperature	62 °C	78 °C
Temperature Rise Above Ambient	40 °C	56 °C

The test is run for 45 minutes, at an ambient temperature of 22 °C. As previously discussed in subsection II-C, the thermistor temperature is a good representation of the HEMT temperature, and so this indicates that the thermal solution has been properly designed and implemented for the intended electrical rating of the inverter. No faults were observed over 20 hours of testing 10 half bridge power modules.

Running the modules at 155 A, 120 V, gives them a power rating of 9.3 kW. Given the outer dimensions are 117 x 62 x 59 mm, each module has a volume of 0.43 litres, which translates to a volumetric power density of 21.7 kW/litre. Each module weighs 259 g, giving a gravimetric power density of 35.9 kW/kg.

While this volumetric power density is comparatively low, simple changes to the layout of the half bridge PCB, as well as increasing the DC link voltage to 140 V, can reduce the module volume to 0.275 litres, increasing the power density

to 37.1 kW/litre. Excluding the gate driver board, as is done on many commercial power modules reduces the volume furthermore down to 0.243 litres, giving a power density of 44.6 kW/litre, comparable to state-of-the-art commercial SiC power modules when operating under the same electrical and thermal conditions.

C. Thermal Imaging Test

Thermal imaging is used to examine static and dynamic current sharing of the HEMTs. A power module is run in the full bridge setup mentioned in the previous subsection without its heat sink and TIM, meaning the temperature of the exposed HEMT can be viewed using an infrared camera.

Fig. 15 is a close up thermal image of a bare power module in a full bridge converter operating at a phase current of 35 A at a switching frequency of 50 kHz. Ambient temperature is 24 °C. The converter is operated until the temperature reaches steady state. The HEMTs are covered in black electrical tape to achieve an emissivity of 0.95 to increase measurement reliability [34].

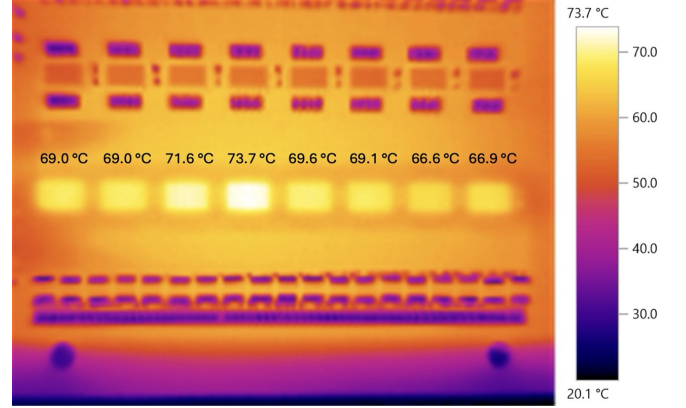


Fig. 15: Thermal image of the top side of the half bridge PCB during the thermal imaging test.

The coldest HEMT has a 42.6 °C temperature rise above ambient, while the hottest HEMT has a 49.7 °C temperature rise above ambient. This means it has a 17% greater temperature rise, which assuming that conduction losses are the dominant loss term means that it is conducting 8% more current, according to

$$T_J - T_{amb} = \frac{P_L}{R_{thermal}} ; \text{ where } P_L = I^2 R_{ds(on)} \quad (4)$$

D. Three-phase Motor Drive Test

To test the functionality of the inverter in a three-phase setup, as well as its ability to recirculate power through a second three-phase inverter, it is connected to the dyno rig discussed previously. LWSR 250-P current sensors measure the phase current allowing field orientated control using MTPA to be implemented via the onboard microcontroller.

Fig. 16 shows the three phase currents of the test inverter driving the test motor in current control mode at a reference

current of 40 A, while the load motor is driven in speed control mode at a speed of 500 rpm. Increasing the switching frequency significantly reduces the ripple current which reduces motor losses, demonstrating one of the key benefits of implementing WBG devices in traction inverters.

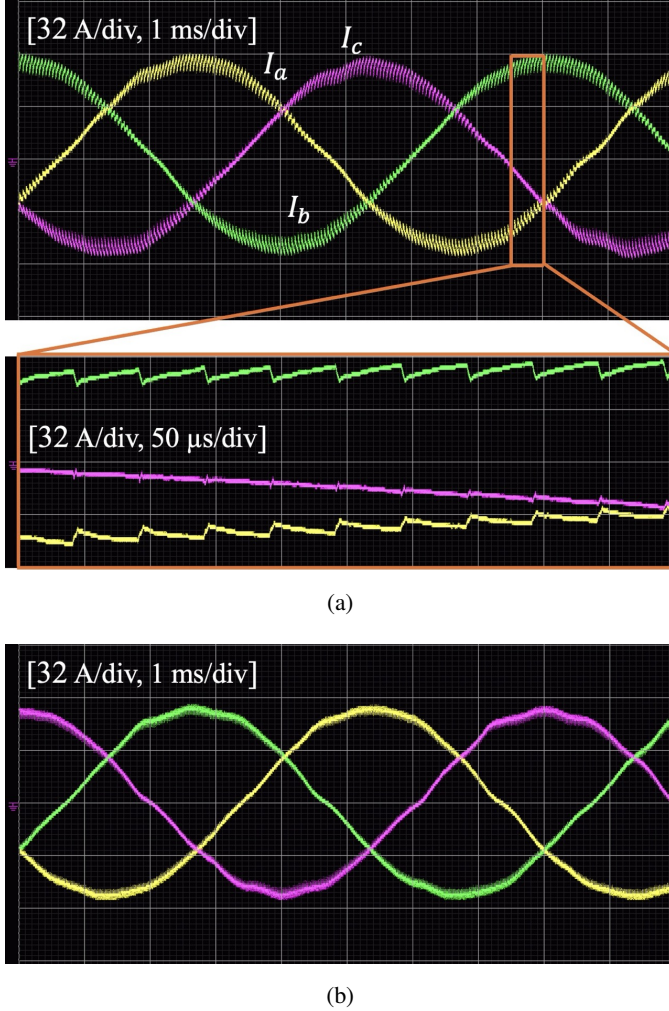


Fig. 16: Oscilloscope waveforms of the three phase currents for two different switching frequencies, a) 20 kHz and b) 50 kHz.

IV. CONCLUSION

This paper presents the design and experimental validation of a dual, three-phase, GaN-based, inverter with eight parallel GaN HEMTs per switch, developed for the characterisation of wide bandgap power electronics in electric vehicle traction applications. The inverter was successfully operated up to a steady state phase current of 155 A at 120 V, for switching frequencies of up to 100 kHz, demonstrating that a high degree of current sharing between parallel HEMTs can be achieved through careful gate drive design and symmetrical layout, as also confirmed by thermal imaging showing comparable device temperatures across all parallel HEMTs. This indicates

that the design can easily operate at switching frequencies far above what is achievable with Si devices. Thermal modelling of the system shows close matching between the measured thermistor temperature and HEMT junction temperature. Operation of the inverter to drive a three-phase motor was successful.

Future work will use the inverter and dyno rig to characterise the efficiency of the inverter and motor across a range of operating conditions, including variations in DC link voltage, switching frequency, and output filter configuration. This will provide a more complete understanding of the system-level benefits and trade-offs associated with the use of GaN HEMTs at high switching frequencies in electric vehicle inverter applications.

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